

# SAT — Secure AI-Enhanced Telemetry

for Edge-Cloud IaaS Infrastructure

---

**Hafsa Metmari**

Master's PFE — FPL Iarache, Abdelmalek Essaâdi University

Supervisor: Prof. Mohammed kouissi

June 2026

[github.com/hafsa-mtm/sat-infra-configs](https://github.com/hafsa-mtm/sat-infra-configs)

# Presentation Plan

**01** Context & Objectives

**02** System Architecture

**03** Security Architecture

**04** DevSecOps & Automation

**05** AI-Based Anomaly Detection

**06** Technical Implementation

**07** Results & Evaluation

**08** Conclusion & Future Work

# 01

## Context & Objectives

General Context • Problem Statement • Research Goals • Methodology

*The convergence of Edge Computing and Cloud IaaS is reshaping enterprise infrastructure — introducing new complexity at every layer.*

## Edge–Cloud IaaS Growth

Enterprises increasingly rely on distributed IaaS environments spanning Edge nodes and centralized Cloud platforms — continuously generating logs, metrics, and security events at scale.

## Operational Complexity

Heterogeneous tools, elastic workloads, and continuous VM lifecycle changes make traditional manual monitoring approaches insufficient and prone to configuration drift.

## Security Attack Surface

Data traversing Edge–Cloud boundaries requires encrypted channels, strict access control, and intelligent behavioral detection — reactive measures are no longer adequate.

*Existing frameworks address these challenges only partially — leaving three critical gaps:*

01

## No Unified Telemetry

Logs, metrics, and security events are collected by separate tools with no unified pipeline — resulting in fragmented visibility and delayed incident response.

02

## No End-to-End Security

Security threats cross the Edge–Cloud boundary with no continuous encryption, no data-layer access control, and no coordinated threat detection across infrastructure layers.

03

## AI in Isolation

AI-based anomaly detection exists only in centralized silos — disconnected from the telemetry pipelines, security events, and automation frameworks that govern the infrastructure it monitors.

# 01 Research Objectives & Methodology

## Research Objectives

01

Unified Edge–Cloud telemetry integration across six parallel pipelines

02

Defense-in-depth security enforcement at network, data, and application layers

03

Intelligent multi-source AI anomaly detection using Isolation Forest

04

DevSecOps automation enabling zero-manual-intervention deployment

## Methodology – DSR

1

Problem Identification & Gap Analysis

2

Research Objective Definition

3

Architecture Design & Formal Modeling

4

Prototype Implementation

5

Empirical Evaluation

6

Contribution Synthesis

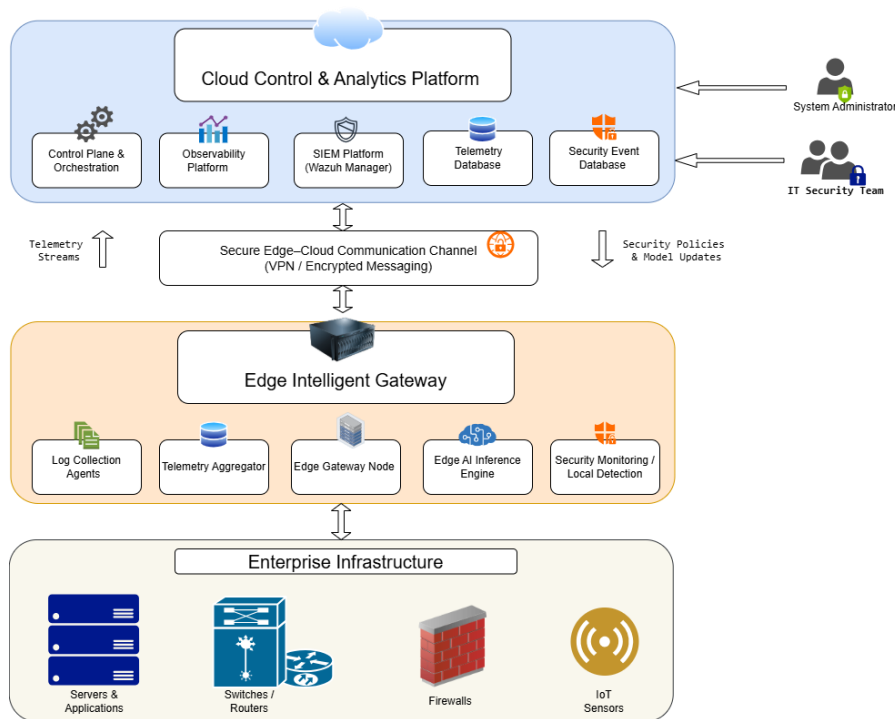
# 02

## System Architecture

High-Level Design • Multi-Edge Scalability

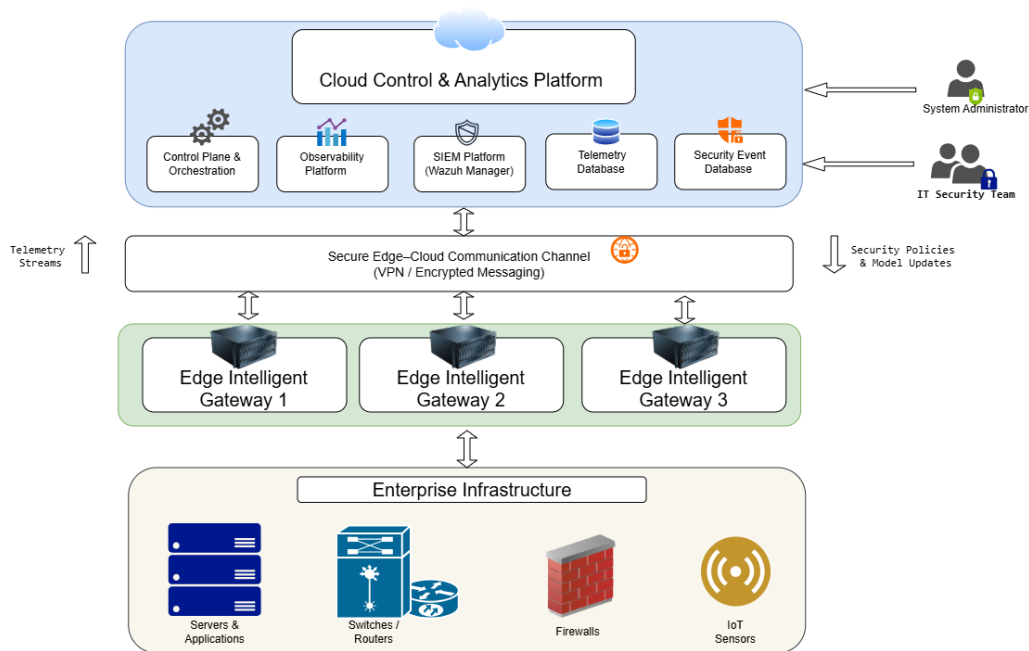
# SAT Platform — High-Level Architecture

*SAT integrates six telemetry pipelines, three security enforcement layers, an AI anomaly detection engine, and a DevSecOps automation framework into one unified platform.*



# Multi-Edge Scalable Deployment Architecture

*Designed for N Edge nodes — validated on 2, scalable to N through Kafka decoupling and automated Ansible onboarding.*



# 03

## Security Architecture

Three-Layer Defense in Depth • Attack Simulation • SIEM Detection

# Three-Layer Defense-in-Depth Security

*Security is enforced at three independent architectural layers — eliminating any single point of failure across the Edge–Cloud boundary.*

## Layer 1 — Network

### WireGuard VPN — ChaCha20-Poly1305 Encryption

0% packet loss | <2ms RTT |  
Continuous tunnel

## Layer 2 — Data

### Elasticsearch X-Pack Authentication

401 on ALL unauthenticated requests |  
Scoped service tokens

## Layer 3 — Application

### Wazuh SIEM — Rule-Based Threat Detection

1,451 alerts | 3 MITRE ATT&CK  
techniques | Real-time

All Kafka telemetry traffic routed through WireGuard VPN — zero plaintext transmission across the Edge–Cloud boundary

# Attack Simulation & SIEM Detection

Four controlled attack scenarios executed on VM2 — Wazuh detected the complete multi-stage attack chain with second-level timestamp granularity.

Documents (1,385)

Field statistics

Columns4

Sort fields1

Exit fullscreen (esc)

| timestamp                                                                                                               | rule.description | rule.level | agent.name |
|-------------------------------------------------------------------------------------------------------------------------|------------------|------------|------------|
| <div><div></div><div>Apr 27, 2026 @ 01:22:33.282</div><div>PAM: Login session closed.</div></div>                       | 3                | edge       |            |
| <div><div></div><div>Apr 26, 2026 @ 22:42:21.261</div><div>PAM: Login session closed.</div></div>                       | 3                | edge       |            |
| <div><div></div><div>Apr 26, 2026 @ 22:42:21.256</div><div>PAM: Login session opened.</div></div>                       | 3                | edge       |            |
| <div><div></div><div>Apr 26, 2026 @ 22:42:21.255</div><div>Successful sudo to ROOT executed.</div></div>                | 3                | edge       |            |
| <div><div></div><div>Apr 26, 2026 @ 22:42:15.384</div><div>PAM: Login session closed.</div></div>                       | 3                | edge       |            |
| <div><div></div><div>Apr 26, 2026 @ 22:42:15.384</div><div>PAM: Login session opened.</div></div>                       | 3                | edge       |            |
| <div><div></div><div>Apr 26, 2026 @ 22:42:15.259</div><div>Successful sudo to ROOT executed.</div></div>                | 3                | edge       |            |
| <div><div></div><div>Apr 26, 2026 @ 22:40:30.878</div><div>sshd: Attempt to login using a non-existent user</div></div> | 5                | edge       |            |
| <div><div></div><div>Apr 26, 2026 @ 22:40:28.876</div><div>PAM: User login failed.</div></div>                          | 5                | edge       |            |
| <div><div></div><div>Apr 26, 2026 @ 22:40:24.872</div><div>sshd: Attempt to login using a non-existent user</div></div> | 5                | edge       |            |

Rows per page: 100

<

1

2

3

4

5

>

# 04

## DevSecOps & Automation

Infrastructure-as-Code • Continuous Monitoring • Version Control

# Infrastructure-as-Code — Three Automation Artifacts

*The complete SAT infrastructure is provisioned through three complementary IaC artifacts — enabling zero-manual-intervention deployment from a version-controlled repository.*

## Kolla-Ansible

OpenStack IaaS Platform

377 Ansible tasks

232 config changes

0 failures

6 OpenStack services

## Docker Compose

Service Stack Definitions

Cloud + Edge stacks

Port mappings & volumes

Restart policies

1 command deployment

## Ansible Role

edge\_monitoring

16 automated tasks

Jinja2 templates

0 failures

Idempotent execution

393 total Ansible tasks executed | 0 failures | All configs version-controlled at [github.com/hafsa-mtm/sat-infra-configs](https://github.com/hafsa-mtm/sat-infra-configs)

# Continuous Monitoring & Version Control

*Autonomous operation across all pipeline components — systemd auto-restart, Docker restart policies, and scheduled cron jobs ensure zero manual intervention.*

## VM2 — Edge Node Automation

- Wazuh Agent — systemd auto-restart
- Prometheus — 15s scraping interval
- Node Exporter — continuous exposition
- 6 Docker containers — restart: unless-stopped
- Filebeat + Metricbeat — auto-recovery

## VM1 — Cloud Node Automation

- 38 containers — restart: unless-stopped
- Wazuh alert ingestion — cron every 5 min
- AI pipeline — scheduled execution
- ELK Stack + Kafka — always-on
- Grafana alerting — 1-min evaluation

# 05

## AI-Based Anomaly Detection

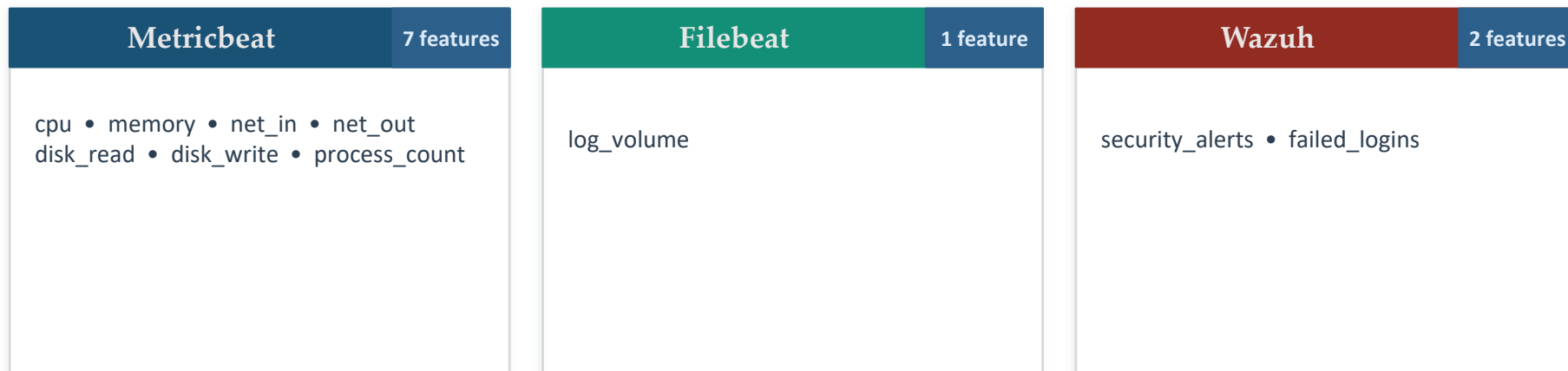
Isolation Forest Pipeline • Multi-Source Fusion • 100% Recall

# Isolation Forest — Multi-Source AI Pipeline

*An unsupervised Isolation Forest model fuses 10 features from 3 independent data sources — enabling detection of 7 anomaly categories without any labeled training data.*



## 10-Dimensional Feature Space



## 05 AI Anomaly Detection — Results

*Hybrid supervised-unsupervised evaluation against 4 controlled attack scenarios — 12 ground-truth records within 1,000 observations.*

100%

**Recall**

All attacks detected  
Zero false negatives

92.5%

**Accuracy**

Overall classification  
correctness

87

**Anomalies**

Detected out of  
1,000 records

8.7%

**Anomaly Rate**

Within expected  
10% contamination

### Score Separation

0.1404 units between normal (avg -0.3594) and anomalous observations — confirming behavioral boundary learning beyond arbitrary classification.

### CPU Stress Attack Detected

CPU spike: 2.89% → 200% (69× increase). Three most anomalous records scored below -0.80, temporally aligned with attack injection timestamps.

# 06

## Technical Implementation

Environment • Deployment • End-to-End Data Flow

# Experimental Environment — VM1 vs VM2

*Two Ubuntu 22.04.5 virtual machines deployed on VMware Workstation 17 Pro — faithfully reproducing the logical topology of a production Edge–Cloud IaaS deployment.*

## VM1 — Cloud / Control Plane

|             |                                                  |
|-------------|--------------------------------------------------|
| Hostname    | cloud                                            |
| RAM / vCPU  | 15 GB / 4 cores                                  |
| IP Address  | 192.168.87.132                                   |
| VPN Address | 10.0.0.1                                         |
| Services    | 38 containerized                                 |
| Key Stack   | OpenStack, ELK, Kafka,<br>Wazuh Manager, Grafana |

—  
WireGuard  
VPN

## VM2 — Edge Gateway Node

|             |                                                  |
|-------------|--------------------------------------------------|
| Hostname    | edge                                             |
| RAM / vCPU  | 7.7 GB / 2 cores                                 |
| IP Address  | 192.168.87.131                                   |
| VPN Address | 10.0.0.2                                         |
| Services    | 6 Docker + 3 systemd                             |
| Key Stack   | Filebeat, Metricbeat,<br>Wazuh Agent, Prometheus |

# Deployment Summary

*The complete SAT stack was deployed autonomously — Cloud IaaS via Kolla-Ansible, Edge stack via Docker Compose, secure channel via WireGuard.*

## Cloud Deployment

- OpenStack 2024.2 via Kolla-Ansible
- 377 tasks — 0 failures — 6 services
- ELK 8.13.0 — Docker Compose
- Wazuh Manager 4.7.0 — Docker
- Kafka 7.5.0 + ZooKeeper — Docker
- Grafana 10.4.0 — Docker

## Edge Deployment

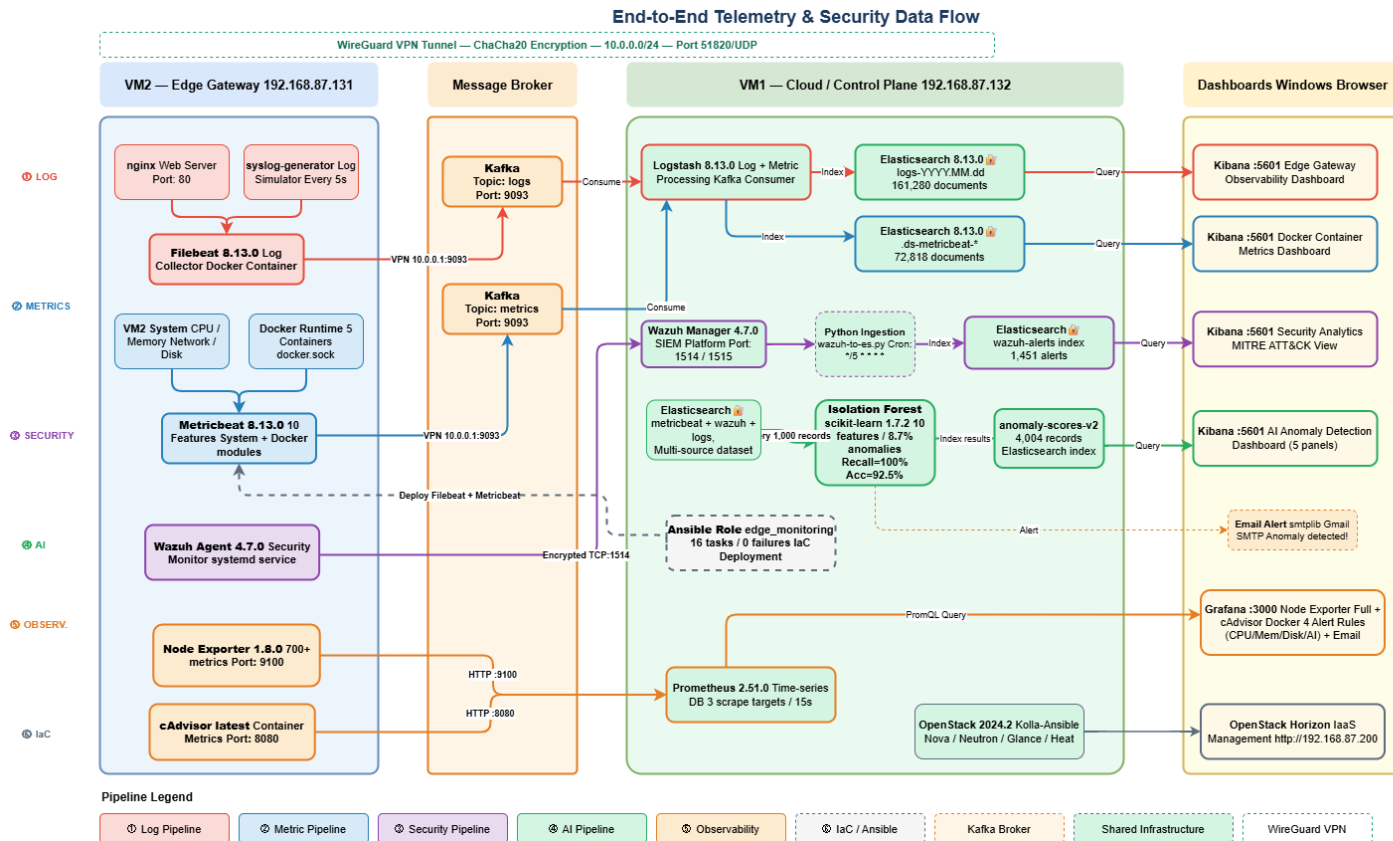
- Docker Engine 29.4.0 + Compose v5.1.3
- Filebeat 8.13.0 — Docker container
- Metricbeat 8.13.0 — Docker container
- Wazuh Agent 4.7.0 — systemd service
- Prometheus 2.51.0 — systemd service
- Node Exporter 1.8.0 — systemd service

## Secure Channel

- WireGuard 1.0.20210914
- Subnet 10.0.0.0/24
- ChaCha20-Poly1305 encryption
- 0% packet loss — <2ms RTT
- Port 51820/UDP — PersistentKeepalive 25s
- systemd auto-start on boot

# End-to-End Telemetry & Security Data Flow

The complete platform in one diagram — every component, every pipeline, every connection. 252,348 documents — 10 days — zero failures.



► **Live Technical Demonstration —  
Kibana | Grafana | Elasticsearch  
| AI Dashboard**

# 07

## Results & Evaluation

Observability • Resource Efficiency • Objectives Validated • Related Works

# Observability Platform — 6 Purpose-Built Dashboards

*Complete four-pillar observability coverage — logs, metrics, security, and AI analytics — across Kibana and Grafana.*

## Kibana — 4 Dashboards

### Edge Gateway Observability

logs-\* + metricbeat-\* | 4 panels — log volume, CPU, memory

### Docker Container Metrics

metricbeat-\* | 2 panels — CPU & memory per container

### Security Analytics

wazuh-alerts (1,451) | 5 panels — MITRE ATT&CK mapping

### AI Anomaly Detection

anomaly-scores-v2 (4,004) | 5 panels — scores, correlation, CPU

## Grafana — 2 Dashboards

### Node Exporter Full

ID 1860 — CPU 2.1%, RAM 13%, Disk 59.8%  
700+ system-level metrics

### cAdvisor Docker

ID 14282 — per-container CPU, memory  
and network throughput

Dual-layer alerting: Grafana SMTP + Python smtplib

# Resource Efficiency — Asymmetric Design Validated

*The architectural principle of concentrating intelligence at the Cloud while maintaining minimal Edge footprint is empirically confirmed.*

## 7.9% RAM — Complete Edge monitoring stack — 6 containers + 3 systemd services

### VM2 Edge Gateway — Per-Service RAM

|                   |               |             |
|-------------------|---------------|-------------|
| Filebeat 8.13.0   | 63 MB         | 0.82%       |
| Metricbeat 8.13.0 | 55 MB         | 0.71%       |
| Wazuh Agent 4.7.0 | 22 MB         | 0.29%       |
| Prometheus 2.51.0 | 22 MB         | 0.29%       |
| Node Exporter     | 8 MB          | 0.11%       |
| nginx + apps      | 14 MB         | 0.18%       |
| <b>TOTAL</b>      | <b>609 MB</b> | <b>7.9%</b> |

### VM1 Cloud Control Plane — RAM Distribution

|                              |                    |
|------------------------------|--------------------|
| Elasticsearch 8.13.0         | ~4 GB              |
| OpenStack (Nova, Neutron...) | ~3 GB              |
| Wazuh Manager 4.7.0          | ~1 GB              |
| Kafka + ZooKeeper            | ~512 MB            |
| Logstash + Kibana            | ~1 GB              |
| Grafana + AI pipeline        | ~512 MB            |
| <b>TOTAL (38 containers)</b> | <b>10 GB / 64%</b> |

# Research Objectives — Fully Validated

*All four research objectives are confirmed by quantitative, reproducible, and independently verifiable empirical evidence.*

**01**

## Unified Edge–Cloud Telemetry Integration

252,348 documents | 6 pipelines | 10 days | 0 failures | 0 data loss incidents

**02**

## Defense-in-Depth Security Enforcement

0% VPN packet loss | <2ms latency | 401 on all unauthenticated requests | 3 MITRE ATT&CK techniques detected

**03**

## Intelligent Multi-Source AI Anomaly Detection

100% Recall | 92.5% Accuracy | 87 anomalies | 10D feature space | 4/4 attack scenarios detected

**04**

## DevSecOps Infrastructure Automation

393 total tasks | 0 failures | 3 IaC artifacts | Public GitHub repo | Zero manual intervention

## 07 Comparison with Related Works

*SAT is the only framework that simultaneously addresses all four architectural dimensions — advancing the state of the art across security, AI, observability, and automation.*

| Dimension             | SAT Platform                  | Best Related Work | SAT Advantage          |
|-----------------------|-------------------------------|-------------------|------------------------|
| Telemetry pipelines   | <b>6 — Kafka-buffered</b>     | 2-3 direct        | Resilient decoupling   |
| Security layers       | <b>3 (VPN+Auth+SIEM)</b>      | 1 (SIEM only)     | Defense-in-depth       |
| AI feature sources    | <b>3 (multi-source)</b>       | 1 (single-source) | Cross-domain detection |
| AI Recall             | <b>100%</b>                   | Not reported      | Quantified performance |
| Observability pillars | <b>4 (complete)</b>           | 1-2 (partial)     | Unified coverage       |
| IaC automation        | <b>3 artifacts, 393 tasks</b> | None reported     | Full reproducibility   |
| Edge RAM overhead     | <b>7.9%</b>                   | Not reported      | Lightweight deployment |

# 08

## Conclusion & Future Work

Contributions • Limitations • Perspectives

# Conclusion & Future Work

## SAT Demonstrates:

- Unified Edge–Cloud telemetry is achievable — 252,348 docs, 0 failures
- Defense-in-depth security adds <2ms overhead — 0% packet loss
- Unsupervised AI delivers 100% recall without labeled training data
- IaC automation enables zero-manual deployment — 393 tasks, 0 failures
- 7.9% Edge RAM — complete monitoring stack deployable on real hardware

[github.com/hafsa-mtm/sat-infra-configs](https://github.com/hafsa-mtm/sat-infra-configs)

## Future Directions:

- 1 Streaming AI inference — Kafka Streams
- 2 Multi-Edge topology validation
- 3 Wazuh active response rules
- 4 Terraform cross-cloud provisioning
- 5 Advanced APT attack scenarios

Thank you — Questions welcome